

Reference audio



Candidate encoding



Measure file bytes $B(c)$

Decode + align

Common grid · fixed delay
Full reference duration

Fidelity evaluation



Global fidelity

Loud frames dominate
energy-weighted global error

$$Q_G \geq g$$

Temporal tail

Worst active-frame relative errors
expose quiet-frame distortion

$$Q_T \geq t$$

Both constraints must pass



Smallest feasible file

$$c^* = \arg \min_{c \text{ feasible}} B(c)$$